

CONTACT INFORMATION	310 N Monroe St 114C Animal Science Building Oklahoma State University Stillwater, Oklahoma 74078 USA	<i>E-mail:</i> samorim@okstate.edu <i>Phone:</i> (405)-744-9397 <i>WWW:</i> https://sabrinaam.github.io/
RESEARCH INTERESTS	I'm an Animal Scientist with an MSc in Genetics and Animal Breeding and a Ph.D. in Animal Sciences (Quantitative Genetics). My research interests include quantitative genetics and its applications in animal breeding. My overarching research interest is understanding the genotype-phenotype map in livestock species using bioinformatics and statistical genetics. I'm interested in better understanding the genetic architecture of economically important traits and applying and developing statistical methods for prediction in the multi-omics era.	
EDUCATION	Virginia Tech , Blacksburg, Virginia USA Ph.D., Animal Sciences, August 2024 • Dissertation: "Genetic Heterogeneity of Residual Variance for Production and Functional Traits in American Angus Cattle." • Advisor: Prof. Dr. Gota Morota • Committee: Drs. James Chen, Jason Holliday, Johan Osorio Universidade Estadual Paulista , Jaboticabal, São Paulo Brazil M.S., Animal Breeding and Genetics, October 2020 • Thesis: "Epistatic interactions associated with fatty acid profile of beef from Nellore cattle." • Advisor: Prof. Dr. Fernando Sebastián Baldi Rey • Committee: Drs. Danísio Prado Munari, Henrique Nunes de Oliveira and Júlio César de Carvalho Balieiro Universidade Federal de Santa Catarina , Florianópolis, Santa Catarina Brazil B.S., Animal Sciences, December 2017 • Thesis: "Application of PCR-RFLP in the Exon 6 region of the IGF-I gene in Crioula Lageana breed animals." • Advisor: Prof. Dr. André Luís Ferreira Lima	
PROFESSIONAL POSITIONS	Department of Animal and Food Sciences Oklahoma State University , Stillwater, Oklahoma USA Assistant Professor of Quantitative Genetics and Genomics Principal Investigator FTE: 70% Research & 30% Teaching	08/2024 - Present

VISITING & TEMPORARY POSITIONS	Laboratory of Quantitative Genetics (Morota Lab) Department of Animal and Poultry Sciences Virginia Tech, Blacksburg, Virginia, USA Masters FAPESP fellowship (BEPE-MS) Host: Dr. Gota Morota	05/2019 - 11/2019
WORK EXPERIENCE	Department of Animal Sciences Virginia Tech, Blacksburg, Virginia USA Graduate Research Assistant Beef Product Development Group Genus - ABS Global, Deforest, Wisconsin USA Beef Genetic Data Analyst (student internship)	08/2021 - 08/2024 05/2023 - 08/2023
	College of Agriculture and Life Sciences Virginia Tech, Blacksburg, Virginia USA Graduate Teaching Scholar	08/2022 - 08/2024
	Departamento de Aquicultura Universidade Federal de Santa Catarina, Florianópolis, Santa Catarina Brazil Administrative Assitant (student intern)	09/2017 - 12/2017
	Departamento de Zootecnia e Desenvolvimento Rural Universidade Federal de Santa Catarina, Florianópolis, Santa Catarina Brazil Undergraduate Research Intern	01/2016 - 07/2017
	Khan Academy Brazil Project Bonita Produções, São Paulo, São Paulo Brazil Translator, Instructor and Producer	02/2014 - 02/2018
PROFESSIONAL SOCIETY MEMBERSHIPS	- American Society of Animal Science. 2024 - Present - North American Colleges and Teachers of Agriculture. 2022 - Present	
AWARDS AND RECOGNITION		
2024	- Graduate Student Teaching Award. NACTA, USA. - Center for Advanced Innovation in Agriculture (CAIA) Big Event Poster Award (Graduate Student Competition) (\$250). Virginia Tech, USA. - Center for Advanced Innovation in Agriculture (CAIA) Graduate Travel Award (\$1,000). CAIA, Virginia Tech, USA.	

- 2023
 - Dwight Williams Scholarship Award (\$1,000). Genus/ABS, USA.
 - Reviewer Recognition. Journal of Animal Science, USA.
- 2022
 - Modern Programming in Genome to Phenome Course Scholarship (\$1,500). University of California (Davis), USA.
- 2018
 - Young Scientist Registration Scholarship (\$1,200). World Congress on Genetics Applied to Live-stock Production, Auckland, New Zealand.

EDITORIAL
ACTIVITIES

GUEST EDITOR

- 2025
 - 1. Beline M, **Amorim ST**, and Reis BR. 2025. Special Issue: Advancing Meat Quality Through Genetic and Nutritional Insights. *Animals*.

AD HOC REVIEWER

Table 1: Summary of manuscripts reviewed

Journal	Number
Animals	2
Archives Animal Breeding	1
Animal Reproduction	1
BMC Genomics	3
Discover Animals	1
Frontiers in Animal Science	0
Frontiers in Veterinary Science	1
IEEE Transactions on AgriFood Electronics	1
Journal of Animal Breeding and Genetics	1
Journal of Animal Science	3
Mammalian Genome	1
Scientific Reports	1
Tropical Animal Health and Production	2
Total	18

Year	Number
2026	3
2025	7
2024	3
2023	3
2022	2
Total	17

PUBLICATIONS

Table 2: Summary of my publications

	First/Corresponding author	Co-Author	Total
Peer-reviewed research journal articles	6	15	21
Peer-reviewed conference proceedings	1	1	2
Outreach publications	0	1	1
Conference abstracts	5	10	15
Total	12	27	39

PEER-REVIEWED
RESEARCH JOURNAL
ARTICLES

2025

21. Morales, A, **Amorim, ST**, Lizana C, Pulido R, Hanigan M, Cockrum R, Morota G. Incorporating heterosis effects enhances genetic evaluations for milk production and functional traits in Chilean crossbred dairy cows. Journal of Dairy Science, 2025. DOI.

20. **Amorim, ST**, Stafuzza NB, Cifuentes DC, Moraes JGN, dos Reis BR, Messmann R, Camaripano L, Baldi F. Genomic Study for Pregnancy Loss in Brahman Cattle. Journal of Animal Science, 2025. DOI.

19. **Amorim, ST**, Retallick K, Garcia A, Ibañez-Escriche N, Morota G. Genetic Heterogeneity of Residual Variance for Foot Score Traits in American Angus Cattle. Journal of Animal Breeding and Genetics, 2025. DOI.

18. Bodmer JS, Beline M, Yen CN, Wicks JC, **Amorim, ST**, Roth EC, Biase FH, Koohmaraie M, Matarneh S, Shi TH, Silva SL, Gerrard DE. In Vitro Proteolysis Mirrors Intact Muscle Maturation in Beef Carcasses. Meat Science, 2025. DOI.

2024

17. Cardona-Cifuentes D, Neira JDR, Albuquerque LG, Espigolan R, Gonzalez-Herrera LG, **Amorim ST**, Lopez-Correa RD, Aguilar I, Baldi F. Influence of variance component estimates on genomic predictions for growth and reproductive-related traits in Nellore cattle. Journal of Animal Breeding and Genetics, 2024. DOI.

16. Temp LB, Brunes LC, Pereira LS, **Amorim ST**, Magnabosco CU, Lobo RB, Brito OC, Viacava R, Baldi F. Effect of genetic and sex effect on genomic prediction for horn development in Nellore cattle. Livestock Science, 2024. DOI.

15. Soares BB, Brunes LC, Baldi FS, Carmo AS, Pereira LS, Carvalho RA, Narciso MG, **Amorim ST**, Sainz RD, Magnabosco CU. Genetic parameters for visual scores, growth and carcass traits in Nellore Cattle. Anais da Academia Brasileira de Ciências, 2024. DOI.

14. Negreiros MP, **Amorim ST**, Lôbo RB, Brunes LC, Magnabosco CU, Bergmann JAG, Espigolan R, Pereira ASC, Baldi F. Genetic correlation estimates between calving ease in primiparous cows and economically important traits in Nellore Cattle. *Journal of Animal Breeding and Genetics*, 2024. [DOI](#).

2023

13. Silva Neto JB, Mota LFM, **Amorim ST**, Peripolli E, Brito LF, Magnabosco CU, Baldi F. Genotype-by-environment interactions for feed efficiency traits in Nellore cattle based on bi-trait reaction norm models. *Genetics Selection Evolution*, 2023. [DOI](#).

2022

12. Kadlec R, Indest S, Castro K, Waqar S, Campos LM, **Amorim ST**, Bi Y, Haningan MD, Morota G. Automated acquisition of top-view dairy cow depth image data using an RGB-D sensor camera. *Translational Animal Science*, 2022. [DOI](#).
11. **Amorim ST**, Tsuyuzaki K, Nikaido I, Morota G. Improved MeSH analysis software tools for farm animals. *Animal Genetics*, 2022. [DOI](#).

2021

10. Tonussi RL, Londoño-Gil M, De Oliveira Silva RM, Magalhães AFB, **Amorim ST**, Kluska S, Espigolan R, Peripolli E, Pereira ASC, Lôbo RB, Aguilar I, Lourenço DAL, Baldi F. Accuracy of genomic breeding values and predictive ability for postweaning liveweight and age at first calving in a Nellore cattle population with missing sire information. *Tropical Animal Health and Production*, 2021. [DOI](#).
9. **Amorim ST**, Stafuzza NB, Kluska S, Peripolli E, Pereira ASC, Muller LF, De Albuquerque LG, Baldi F. Genome-wide interaction study reveals epistatic interactions for beef lipid-related traits in Nellore cattle. *Animal Genetics*, 2021. [DOI](#).
8. Berton MP, Lemos M, Chud T, Stafuzza NB, Kluska S, **Amorim ST**, Lopes LSF, Pereira ASC, Bickhart D, Liu G, Albuquerque LG, Baldi F. Genome-wide association study between copy number variation regions and carcass and meat quality traits in Nellore cattle. *Animal Production Science*, 2021. [DOI](#).
7. Feitosa FLB, Pereira ASC, Mueller L, Fonseca P, Braz C, **Amorim ST**, Espigolan R, Lemos MVA, Albuquerque LG, Schenkel F, Brito L, Stafuzza NB, Baldi F. Genome-wide association study for beef fatty acid profile using haplotypes in Nellore cattle. *Livestock Science*, 2021. [DOI](#).

2020

6. **Amorim ST**, Yu H, Momen M, Albuquerque LG, Pereira ASC, Baldi F, Morota G. An assessment of genomic connectedness measures in Nellore cattle. *Journal of Animal Science*, 2020. [DOI](#).

2019

5. Feitosa FLB, Pereira ASC, **Amorim ST**, Peripolli E, Silva RMO, Braz CU, Ferrinho AM, Schenkel FS, Brito LF, Espigolan R, De Albuquerque LG, Baldi F. Comparison between haplotype-based and individual SNP-based genomic predictions for beef fatty acid profile in Nellore cattle. *Journal of Animal Breeding and Genetics*, 2019. [DOI](#).

4. Peripolli E, Stafuzza NB, **Amorim ST**, Lemos MVA, Grigoletto L, Kluska S, Ferraz JBS, Eler JP, Mattos EC, Baldi F. Genome-wide scan for runs of homozygosity in composite Montana Tropical® beef cattle. *Journal of Animal Breeding and Genetics*, 2019. [DOI](#).
3. Stivanin T, Maia FMC, Migliorini E, Kluska S, **Amorim ST**, Lovatto FS, Martins EN. Evaluation of selection criteria in laying quails (*Coturnix coturnix japonica*). *Livestock Research for Rural Development*, 2019. [DOI](#).

2018

2. Bonamy M, Kluska S, Peripolli E, Lemos MVA, **Amorim ST**, Vaca RJ, Lobo RB, Castro LM, Faria CU, Ferrari FB, Baldi F. Genetic association between different criteria to define sexual precocious heifers with growth, carcass, reproductive, and feed efficiency indicator traits in Nellore cattle using genomic information. *Journal of Animal Breeding and Genetics*, 2018. [DOI](#).
1. **Amorim ST**, Kluska S, Berton MP, Lemos MVA, Peripolli E, Stafuzza NB, Martin JF, Alvarez MS, Gavina BV, Toro MA, Banchemo G, Oliveira PS, Grigoletto L, Eler JP, Baldi F, Ferraz JBS. Genomic study for maternal related traits in Santa Inês sheep breed. *Livestock Science*, 2018. [DOI](#).

PEER-REVIEWED
CONFERENCE
PROCEEDINGS

2018

2. **Amorim ST**, Eler JP, Grigoletto L, Lemos MVA, Baldi F, Ferraz JBS. Genome Wide Association Study (GWAS) for body weight traits in Santa Ines Sheep breed. In: *World Congress On Genetics Applied To Livestock Production*, 2018, Auckland. *Proceedings Of The World Congress On Genetics Applied To Livestock Production, Volume Genetic Gain - In Challenging Environments*, 2018. V. 11. P. 708-708.
1. Berton M, **Amorim ST**, Kluska S, Lemos MVA, Olivieri B, Tonussi R, Peripolli E, Eler JP, Baldi F, Ferraz JBS. Genetic Parameters and Genome Wide Association Study (GWAS) for maternal related traits in Santa Ines Sheep breed. In: *World Congress On Genetics Applied To Livestock Production*, 2018, Auckland. *Proceedings Of The World Congress On Genetics Applied To Livestock Production, Volume Electronic Poster Session - Species - Ovine*, 2018. V. 11. P. 826-826.

OUTREACH
PUBLICATIONS

2023

1. Smilnak D, **Amorim ST**, Day J, Powell M, Mullins E, Scherer H. Agricultural Cyberbiosecurity: Biotechnology. *Agricultural Cyberbiosecurity Reference Guide*, 2023. [DOI](#).

- 2025
15. Hartman J, **Amorim ST**. Exploring the Genetic Basis of Meat Fatty Acid Composition in Beef Cattle Using MeSH Enrichment Analysis. In: *Freshmen Research Symposium*, Stillwater, OK, 2025.
 14. Bis FC, Carvalho FE, Almeida CA, Reis HB, **Amorim ST**, Stafuzza NB, Oliveira ECM, Negreiros MP, Ferraz JB, Baldi F. Estudo de associação genômica ampla ponderado de passo único para perda gestacional em novilhas Nelore. In: *XVI SBMA*, Piracicaba, SP, Brazil, 2025.
 13. Reis BR, Silva G, Marcon H, Mazeres KR, Ferreira, MFL, **Amorim ST**, Vedovatto M, Beline M. Impact of shade and chromium supplementation on performance and blood metabolites of finishing steers under heat stress conditions. In: *2025 ASAS-CASAS Annual Meeting*, Hollywood, FL, 2025.
 12. **Amorim ST**, Stafuzza NB, Cifuentes DC, Moraes JGN, Reis BR, Messmann R, Camaripano L, Baldi F. Genomic Study for Pregnancy Loss in Brahman Cattle. In: *2025 ASAS-CASAS Annual Meeting*, Hollywood, FL, 2025.
- 2022
11. Bi Y, Kadlec R, Castro K, Indest S, **Amorim ST**, Morota G. Development of Automated Computer Vision Systems for Evaluating Dairy Cow Body Weight and Body Condition Scores. In: The American Society of Animal Science. *Journal of Animal Science*, Volume 100, Issue Supplement_3, 2022.
- 2017
10. Leite AA, **Amorim ST**, Rosa VL, Lemes M, Francisco L, Hauptli L, Irgang R, Lima ALF. Exon IV Polymorphism Of Prolactin Gene On Crioulo Lageano Cattle. In: 54^a Reunião Anual da SBZ, 2017, Foz do Iguaçu. *A New View of Animal Science: Challenges and Perspectives*, 2017.
 9. **Amorim ST**, Leite AA, Lemes M, Francisco L, Chiarelli AC, Pereira MC, Dalhke F, Lima ALF. Genetic Monomorphism Of Insulin-Like Growth Hormone On Crioulo Lageano Cattle (*Bos taurus*). In: 54^a Reunião Anual da SBZ, 2017, Foz do Iguaçu. *A New View of Animal Science: Challenges and Perspectives*, 2017.
 8. Rosa VL, **Amorim ST**, Chiarelli AC, Leite AA, Carvalho SRST, Quadros SAF, Oliveira PP, Lima ALF. Identification Of Polymorphisms On The IGF-I Gene In Crioulo Lageano Cattle (*Bos taurus*). In: 54^a Reunião Anual da SBZ, 2017, Foz do Iguaçu. *A New View of Animal Science: Challenges and Perspectives*, 2017.
 7. Dalhke F, Dahmer AP, Oliveira PP, **Amorim ST**, Hauptli L, Lima ALF. Evaluation Of The Influence Of Astaxanthin Inclusion In Feed For Quails (*Coturnix coturnix japonica*) On Performance And Egg Quality Traits. In: 54^a Reunião Anual da SBZ, 2017, Foz do Iguaçu, 2017.
 6. Oliveira RN, Machado D, Pereira MC, Correa Junior AC, **Amorim ST**, Carvalho SRST, Quadros SAF, Lima ALF. Association Between Productive Performance And Gastrointesti-

nal Nematodes Infection In Ile De France Lambs. In: 54th Reunião Anual da SBZ, 2017, Foz do Iguaçu. *A New View of Animal Science: Challenges and Perspectives*, 2017.

5. Wagner B, **Amorim ST**, Pereira TC, Irgang R, Lima ALF, Pereira MC, Quadros SAF, Carvalho SRST. Beef Consumers Preferences: An Approach On The Cities Of Florianopolis And Botucatu. In: 54th Reunião Anual da SBZ, 2017, Foz do Iguaçu. *A New View of Animal Science: Challenges and Perspectives*, 2017.
4. Xavier AC, Machado D, **Amorim ST**, Tavares S, Irgang R, Hauptli L. Efficiency Of The Questionnaire Application Classic Swine Fever Situation In Terms Of Risks By Municipality In Santa Catarina. In: 54th Reunião Anual da SBZ, 2017, Foz do Iguaçu. *A New View of Animal Science: Challenges and Perspectives*, 2017.
3. **Amorim ST**, Lima ALF, Lemes M, Irgang R, Pereira MC, Chiarelli AC. Efficiency in Feed Conversion in Pigs: Association of Favorable Alleles with Nutritional Modeling. In: *Iniciation to Scientific Research Seminar*, Florianópolis, 2017.
2. Machado D, Mallmann DH, Erpen JG, **Amorim ST**, Selig PM, Marian P, Oliveira PP, Hosrtmann R. Knowledge Intensive Livestock: Knowledge Management Maturity Model Applied On Dairy Farming In Santa Catarina. In: 54th Reunião Anual da SBZ, 2017, Foz do Iguaçu. *A New View of Animal Science: Challenges and Perspectives*, 2017.
1. **Amorim ST**, Lima ALF, Bento LL, Leite AA, Pereira TC, Pacheco L, Pereira MC, Irgang R. Isolation and Application of RFLP-PCR on Ghrelin Gene of Swine. In: *Iniciation to Scientific Research Seminar*, Florianópolis, 2017.

INVITED

PRESENTATIONS

4 domestic and 4 international

- | | |
|------|--|
| 2025 | 8. Genomic study for pregnancy loss in Brahman cattle. ASAS–CSAS Annual Meeting, Hollywood, FL, USA. |
| 2022 | 7. Implementation of Medical Subject Headings (MeSH) analysis in farm animals. Animal Sciences Seminar. Blacksburg, VA, USA. |
| | 6. Collaborative Innovation in Agriculture. Center for Advanced Innovation in Agriculture (CAIA) Big Event. Blacksburg, VA, USA. |
| 2020 | 5. The Faces of Animal Sciences. I SEMAZOOT Online. Florianópolis, Santa Catarina, Brazil. |
| 2019 | 4. Genomic connectedness in Nellore beef cattle. NCERA-225 Meeting for Implementation and Strategies for National Beef Cattle Genetic Evaluation. Blacksburg, VA, USA. |
| 2018 | 3. Genome Wide Association Study (GWAS) For Body Weight Traits In Santa Ines Sheep Breed. World Congress on Genetics Applied to Livestock Production. Auckland, New Zealand. |

2. Efficiency in feed conversion in pigs: Association of favorable alleles with nutritional modeling. Scientific Initiation Seminar. Florianópolis, Brazil.
1. Genetic Monomorphism Of Insulin-Like Growth Hormone On Crioulo Lageano Cattle (Bos taurus). Scientific Initiation Seminar. Florianópolis, Brazil.

TEACHING

Oklahoma State University, Stillwater, Oklahoma, USA

Primary Instructor

- ANSI 3433 Animal Breeding **Spring, 2026**
120 participants

Co-Instructor

- ANSI 3433 Animal Breeding **Fall, 2025**
145 participants

Virginia Polytechnic Institute and State University, Blacksburg, Virginia, USA

Primary Instructor

- ALS 3104 Animal Breeding and Genetics [[WWW](#)] **Spring, 2024**
118 participants

Teaching Assistant

- ALS 3104 Animal Breeding and Genetics [[WWW](#)] **Spring, 2023**
115 participants

Teaching Assistant

- ALS 3104 Animal Breeding and Genetics [[WWW](#)] **Spring, 2022**
91 participants

Faculdades Associadas de Uberaba - FAZU, Uberaba, Minas Gerais, Brazil

Lead Instructor

- Genetic Improvement of Beef Cattle Program - Selection, Mating and Crossbreeding Systems [[WWW](#)] **Fall, 2022**
30 participants
- Genetic Improvement of Beef Cattle Program - Selection, Mating and Crossbreeding Systems [[WWW](#)] **Summer, 2022**
33 participants
- Genetic Improvement of Beef Cattle Program - Selection, Mating and Crossbreeding Systems [[WWW](#)] **Summer, 2021**
25 participants
- Genetic Improvement of Beef Cattle Program - Molecular Markers [[WWW](#)] **Spring, 2021**

30 participants

- Genetic Improvement of Beef Cattle Program - Selection, Mating and Crossbreeding Systems [WWW] **Fall, 2020**
30 participants

Universidade Estadual Paulista, Jaboticabal, São Paulo, Brazil

Teaching Assistant

- Animal Breeding and Genetics [WWW] **Fall, 2018**
100 participants
- Animal Breeding and Genetics {WWW} **Spring, 2018**
108 participants

Universidade Federal de Santa Catarina, Florianópolis, Santa Catarina, Brazil

Undergraduate Teaching Assistant

- Genetics for Animal Sciences [WWW] **Spring, 2015**
30 participants
- Genetics for Animal Sciences [WWW] **Fall, 2015**
33 participants
- Genetics for Animal Sciences [WWW] **Spring, 2016**
35 participants
- Genetics for Animal Sciences [WWW] **Fall, 2016**
30 participants

Khan Academy Brazil Project, São Paulo, São Paulo, Brazil

Instructor

- Chemistry [WWW] **2015-2018**
Online Content
- Biology [WWW] **2015-2018**
Online Content
- Physics [WWW] **2015-2018**
Online Content
- Calculus [WWW] **2015-2018**
Online Content

Guest Instructor

- **Mississippi State University, Starkville, Mississippi, USA**

ADS 4813/6813- Dairy Farm Management. **Fall, 2025**
Principles of Genetic Evaluations in Dairy Cattle

– **Oklahoma State University**, Stillwater, Oklahoma, USA
ANSI 2111 - Animal and Food Science Professional Development. **Fall, 2025**
Faculty Panel

– **Oklahoma State University**, Stillwater, Oklahoma, USA
ANSI 3433 - Animal Breeding. **Spring, 2025**
Principles of Genetic Evaluations

– **Virginia Tech**, Blacksburg, Virginia, USA
DASC4475 / 4476 - Dairy Enterprise Management. **Spring, 2022**
Principles of Genomic Evaluation

RESEARCH SUPPORT

External Funding

- São Paulo Research Foundation (FAPESP) - \$11,357.00 São Paulo Research Foundation
PI: Fernando Baldi **May, 2019 - November, 2019**
Proposal: An assessment of the relationship between Genomic Connectedness and Prediction Accuracy for fat composition traits in Nellore beef cattle
Role: Co-Principal Investigator
- São Paulo Research Foundation (FAPESP) - \$6,774.00 São Paulo Research Foundation
PI: Fernando Baldi **January, 2019 - October, 2020**
Proposal: Epistatic interactions associated with fatty acid profile of beef from Nellore cattle
Role: Co-Principal Investigator
- São Paulo Research Foundation (FAPESP) - \$16,357.00 São Paulo Research Foundation
PI: Fernando Baldi **December, 2025 - January, 2027**
Proposal: Incorporation of phenotypic causal networks in genome-wide association studies for pregnancy loss in precocious heifers using mixed-effects structural equation models
Role: Co-Principal Investigator

Intramural Funding

- Ferguson College of Agriculture/OAES - \$2,000.00 Oklahoma State University
PI: Sabrina T. Amorim **2024**
Undergraduate Research Scholar Support Program.
Role: Principal Investigator, Undergraduate Research Advisor

ADVISEES AND
TRAINEES

Undergraduate Students

1. Jackson Hartman [[WWW](#)] **09/2024 - 05/2025**

M.S. Students

1. Meire Luiza Wirth (jointly with Fernando Baldi) [[WWW](#)] **10/01/2025 - Present**
2. Flavia Cristina Bis (jointly with Fernando Baldi) [[WWW](#)] **10/01/2025 - Present**

Visiting Scholars

1. Flávia Petra Vieira, Universidade de São Paulo (USP), [[WWW](#)] **08/2025 - 12/2025**

THESIS
COMMITTEES

Ph.D Thesis Committees

1. Makenna McKendrick **2024-Present**
Department of Animal and Food Sciences, Oklahoma State University
Major advisor: Darren Hagen
2. Caitlin Moulder **2025-Present**
Department of Animal and Food Sciences, Oklahoma State University
Major advisor: Darren Hagen
2. Joao Gabriel Barbosa da Silva Neto **2025-Present**
Universidade Estadual Paulista (UNESP), Jaboticabal, Brazil
Major advisors: Fernando Baldi and Luiz Brito

M.S. Thesis Committees

1. Chloe Krause **2025-Present**
Department of Animal and Food Sciences, Oklahoma State University
Major advisor: David Lalman
2. Carson Atwood **2025-Present**
Department of Animal and Food Sciences, Oklahoma State University
Major advisor: Meaghan Meyer
3. Gabriel Gubiani **2025-Present**
Universidade Estadual Paulista (UNESP), Jaboticabal, Brazil
Major advisor: Dr. Fernando Baldi

Professional Specialization Thesis Committees

2. Débora Cristina de Oliveira Martins **2021**
Genetic Improvement of Beef Cattle Program - Faculdades Associadas de Uberaba
Major advisor: Thiago Braga
1. Jéssica Huescar dos Santos **2021**
Genetic Improvement of Beef Cattle Program - Faculdades Associadas de Uberaba
Major advisor: Camila Raymundo

BS Thesis Committees

- | | |
|--|-------------|
| 1. Matheus von Roun Silva | 2025 |
| BS in Veterinary Sciences - Universidade de Sao Paulo (FMVZ-USP) | |
| Major advisor: Angelica Simone Cravo Pereira | |

VISITORS HOSTED

- | | |
|---|----------------------|
| • Anna Carolina Denicol, University of California-Davis | November 2025 |
|---|----------------------|

SERVICE ACTIVITIES

Multistate research activities

- | | |
|--|-----------------------|
| • S1093: Management systems for beef cattle reared in subtropical and tropical environments.
Oklahoma State University representative | 2025 - Present |
| • S1086: Genetic aspects of beef cow adaptation to diverse U.S. production environments
Oklahoma State University representative | 2024 - Present |

National

- | | |
|---|---------------------|
| • Annual Southern Section Meeting Program Committee - Breeding and Genetics
American Society of Animal Science | 2024-Present |
|---|---------------------|

Departmental

- | | |
|--|---------------------|
| • Research Committee
Department of Animal and Food Sciences
Oklahoma State University | 2024–Present |
| • Teaching Committee
Department of Animal and Food Sciences
Oklahoma State University | 2025–Present |
| • Social Committee - Chair
Department of Animal and Food Sciences
Oklahoma State University | 2025–Present |
| • AFS Burger Bash Organizer
Department of Animal and Food Sciences
Oklahoma State University | Summer 2025 |
| • Animal and Food Science Graduate Student Association (AFSGSA)
Academic Coordinator
Department of Animal and Food Sciences
Oklahoma State University | 2025–Present |

Research Area

- AFS Research Day Organizer
Department of Animal and Food Sciences,
Oklahoma State University **Summer 2025**
- Genetics Seminar Co-Instructor
Department of Animal and Food Sciences, Oklahoma State University **Spring 2025**
- Genetics Seminar Co-Instructor
Department of Animal and Food Sciences, Oklahoma State University **Fall 2024**

SOFTWARE TOOLS

Image/Video analysis

- [Automated depth data collection](#)

Bioconductor packages

- [meshr](#)

Github

- <https://github.com/sabrinaam>

PARTICIPATION IN MEETINGS, SYMPOSIUMS, AND WORKSHOPS

- | | |
|------|--|
| 2025 | <ul style="list-style-type: none">• Cattle over Coffee Summer Webinar Series: Management Strategies for Beef Cattle Systems: Growth Promoting Implants. Mississippi State University, USA. Online Event.• 2025 ASAS-CSAS Annual Meeting. Hollywood, FL, USA.• A Single-Step Toward the Future Symposium. University of Georgia, Athens, GA, USA. |
| 2024 | <ul style="list-style-type: none">• Write Winning Grant Proposals Workshop. Stillwater, OK, USA.• Advances in Genome Biology and Technology (AGBT). Phoenix, AZ, USA.• Center for Advanced Innovation in Agriculture (CAIA) Big Event. Blacksburg, VA, USA.• Conference on Higher Education Pedagogy. Blacksburg, VA, USA. |
| 2023 | <ul style="list-style-type: none">• Virginia Tech CALS Career Fair (Representing Genus/ABS Global). Blacksburg, VA, USA. |

	• Wisconsin State FFA Convention. Madison, WI, USA. • Industrial and Systems Engineering Symposium. Blacksburg, VA, USA. • Conference on Higher Education Pedagogy. Blacksburg, VA, USA.
2022	• Industrial and Systems Engineering Symposium. Blacksburg, VA, USA. • Center for Advanced Innovation in Agriculture (CAIA)/Commonwealth Cyber Initiative Southwest Virginia (CCI SWVA) Agricultural Cyber Field Day. Virginia Tech, Blacksburg, VA, USA. April 28. • Center for Advanced Innovation in Agriculture (CAIA) Big Event. Blacksburg, VA, USA.
2021	• Empowering Farmers with Affordable Digital Agricultural Solutions. Blacksburg, VA, USA.
2020	• I Virtual Meeting on Innovations in Animal Production. Online Meeting. • II Symposium on Conservation and Sustainable Management of Domestic Breeds in Danger. Online Meeting. • I COICTA - International Online Congress of Food Science and Technology - Food Security: Strategies in the Pandemic Period. Online Meeting.
2019	• NCERA-225 Meeting. Implementation and Strategies for National Beef Cattle Genetic Evaluation. Christiansburg, VA, USA.
2018	• World Congress on Genetics Applied to Livestock Production. Auckland, New Zealand.
2017	• 54th Annual Meeting of the Brazilian Society of Animal Science. Foz do Iguaçu, Brazil. • Meat Inspection and Quality. Florianópolis, Brazil. • Workshop: Quality Brands in Meats - Red Cytod Marcarne. Foz do Iguaçu, Brazil.
2016	• XXVI Brazilian Congress of Animal Science. Santa Maria, Brazil.
ADDITIONAL TRAINING	
2025	• Peer Coaching of Instruction Program, Oklahoma State University.
2023	• Engaging All Learners with Formative Assessment Tools, Virginia Tech.

- 2022
- Modern Programming in Genome to Phenome, University of California-Davis. Taught by Drs. Hao Cheng, Rohan Fernando and James Chen.
 - Programming and computer algorithms in animal breeding with a focus on single-step GBLUP and genomic selection in practice, University of Georgia. Taught by Dr. Ignacy Misztal, Daniela Lourenco, Andres Legarra and Zulma Vitezica.
- 2021
- Basic Responsible Conduct of Research Course, CITI Program.
 - Understanding and Visualizing Data with Python, University of Michigan.
 - Python for Everybody Specialization, University of Michigan.
 - Research Methods: Introductory Concepts, Instituto Federal de Educação, Ciência e Tecnologia do Rio Grande do Sul. Credit Hours: 30h
 - Knowledge Discovery in Databases: Fundamentals, Instituto Federal de Educação, Ciência e Tecnologia do Rio Grande do Sul. Credit Hours: 20h
 - Agriculture and Sustainable Rural Development, Instituto Federal de Educação, Ciência e Tecnologia do Rio Grande do Sul. Credit Hours: 40h
- 2020
- Python for Genomic Data Science, Johns Hopkins University.
 - Introduction to Genomic Technologies, Johns Hopkins University.
 - Extension in Biology, Instituto Federal de Educação, Ciência e Tecnologia do Rio Grande do Sul. Credit Hours: 40h
 - Extension in Statistics: Concepts and Representations, Instituto Federal de Educação, Ciência e Tecnologia do Rio Grande do Sul. Credit Hours: 20h
 - Extension in Statistics: Measures of Position and Dispersion, Instituto Federal de Educação, Ciência e Tecnologia do Rio Grande do Sul. Credit Hours: 20h
 - Extension in Databases: Fundamentals, Instituto Federal de Educação, Ciência e Tecnologia do Rio Grande do Sul. Credit Hours: 20h
 - Extension in Environmental Education, Instituto Federal de Educação, Ciência e Tecnologia do Rio Grande do Sul. Credit Hours: 40h
 - Extension in Hygiene and Food Quality Control, Instituto Federal de Educação, Ciência e Tecnologia do Rio Grande do Sul. Credit Hours: 50h
- 2019
- GWAS Workshop, Virginia Polytechnic Institute and State University. Credit Hours: 30h
 - Special Topics in Bioinformatics: Genome Alignment, Universidade Estadual Paulista. Credit Hours: 30h

- 2018
 - Special Topics in Genome Sequencing for Genomic Studies, Universidade Estadual Paulista. Credit Hours: 20h
- 2017
 - Management and Use of Poultry and Mammals in Research, Universidade Federal de Santa Catarina. Credit Hours: 17h
- MISCELLANEOUS
 - Languages: English and Portuguese
 - Computer skills
 - Statistical/Numerical computational tools: R and Octave
 - Computer vision and image processing: Python and R
 - Content-description languages: $\text{\LaTeX}$, and Markdown
 - Operating systems: Linux, Windows, and Mac OS X
 - Courses taken for credits at Virginia Tech
 - Spring 2024
 - ALCE 6426: University Teaching Development (Tracy Rutherford)
 - Fall 2023
 - ALCE 6425: University Teaching Development (Tracy Rutherford)
 - Spring 2023
 - ALCE 6416: Intro to Graduate Teaching Scholar (Tracy Rutherford)
 - GRAD 5212: Diversity for a Global Society (Shernita Lee)
 - Fall 2022
 - ALCE 6415: Intro to Graduate Teaching Scholar (Tracy Rutherford)
 - APSC 5014: Professional Development for Graduate Students (Jim Knight)
 - ECE 5554: Computer Vision (Lynn Abbott)
 - Spring 2022
 - FREC 5164: Population Genomics (Jason Holliday)
 - STAT 5616: Statistics in Research (Anne Ryan Driscoll)
 - Fall 2021
 - ALS 5224: Intro to Genomic Data Science (Song Li)
 - STAT 5615: Statistics in Research (Anne Ryan Driscoll)
 - GRAD 5004: GTA Training
 - Courses taken for credits at Universidade Estadual Paulista
 - Fall 2018
 - Introduction to Bayesian Inference applied to Animal Breeding (Henrique Nunes de Oliveira)
 - Statistical Modelling in Genetics and Animal Breeding (Guilherme J. Rosa)

- Principles of Association and Genomic Selection applied to Animal Breeding (Fernando Sebastián Baldi Rey)
- Spring 2018
 - Quantitative Genetics I (Danísio Prado Munari)
 - Meat Genetics (Luis Artur Loyola Chardulo)
 - Selection Index and Introduction to Mixed Models (Lúcia Galvão de Albuquerque)

REFERENCES

References and additional information available upon request.